

תכנון ספרתי מתקדם מעבדה 2

מגישים:

אלעד אסבג – 318530482

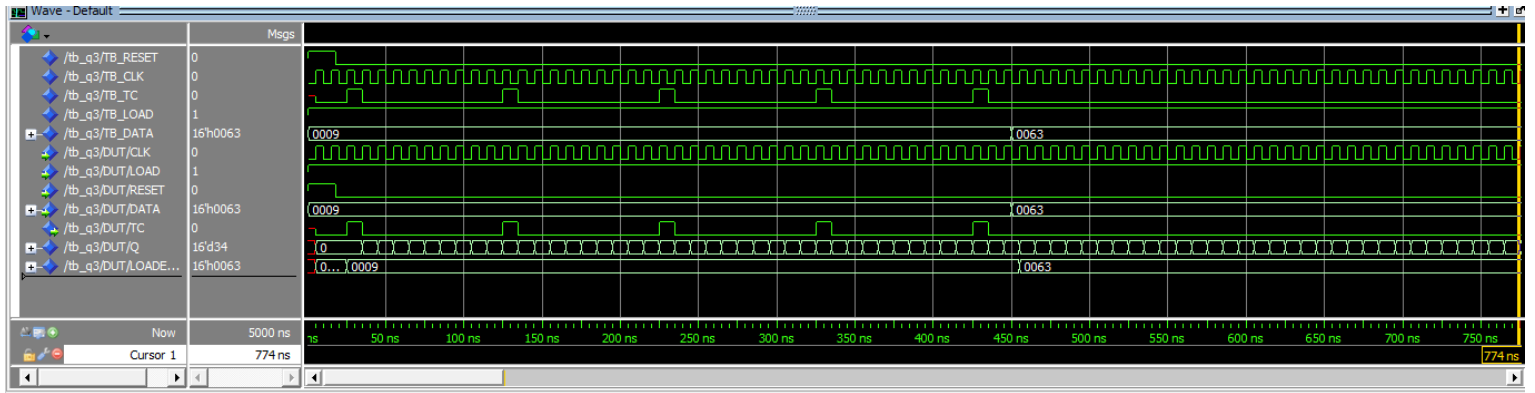
יוסי ברים- 315512848

תכנון התוכנית , ספסל הבדיקה -TB

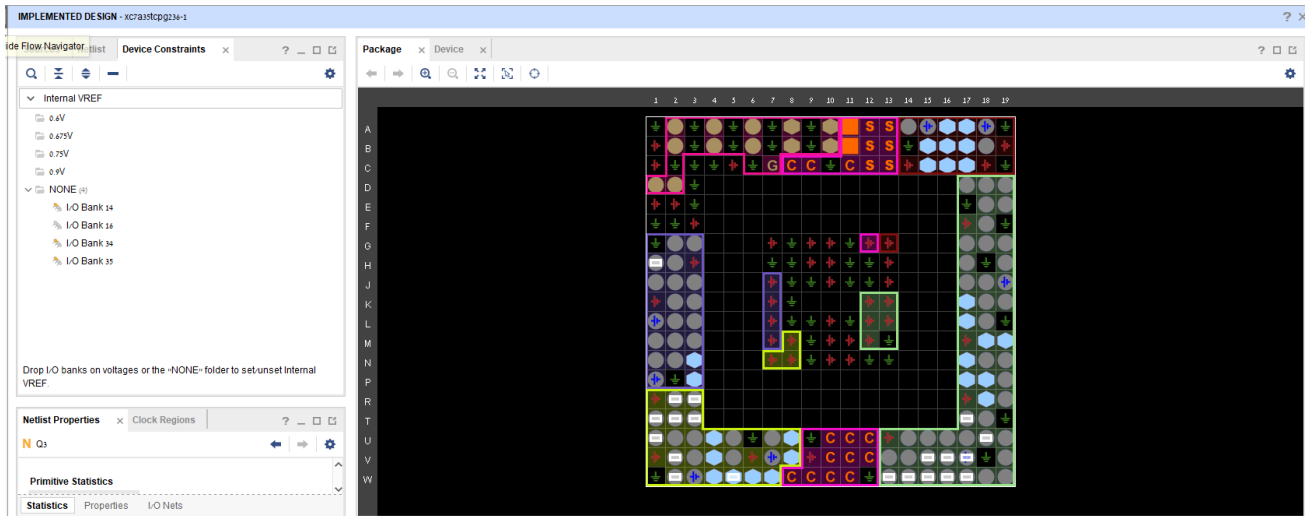
```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3  use IEEE.STD_LOGIC_UNSIGNED.ALL;
4
5  entity TB_Q3 is
6
7  end TB_Q3;
8
9  architecture TB_AAA of TB_Q3 is
10 component Q3
11 port(
12 CLK : in std_logic;
13 RESET : in std_logic;
14 LOAD : in std_logic;
15 DATA : in std_logic_VECTOR(15 downto 0);
16 TC : out std_logic
17 );
18 end component;
19
20 signal TB_RESET: std_logic;
21 signal TB_CLK: std_logic:= '0';
22 signal TB_TC: std_logic;
23 signal TB_LOAD: std_logic;
24 signal TB_DATA : std_logic_VECTOR(15 downto 0);
25
26 begin
27
28 DUT : Q3
29 PORT map(
30 CLK => TB_CLK,
31 RESET => TB_RESET,
32 TC => TB_TC,
33 LOAD => TB_LOAD,
34 DATA => TB_DATA
35 );
36
37
38 L1: TB_RESET <= '1',
39      '0' after 18 ns;
40
41 L2: TB_DATA <= x"0009",
42      x"0063" after 450 ns;
43 L3: TB_LOAD <= '1';
44 L2: TB_CLK <= not (TB_CLK) after 5 ns; -- בסביל שהקלט יפלט כל שניה
45 end TB_AAA;
```

```
1
2  library IEEE;
3  use IEEE.STD_LOGIC_1164.ALL;
4  use IEEE.STD_LOGIC_UNSIGNED.ALL;
5
6  entity Q3 is
7  port(
8  CLK : in std_logic;
9  LOAD : in std_logic;
10 RESET : in std_logic;
11 DATA : in std_logic_VECTOR(15 downto 0);
12 TC : out std_logic
13 );
14 end Q3;
15
16 architecture AAA of Q3 is
17 signal Q : std_logic_VECTOR(15 downto 0);
18 signal LOADED_DATA : std_logic_VECTOR(15 downto 0);
19
20 begin
21
22 process
23 begin
24 wait until rising_edge(CLK);
25 if (RESET = '1') then
26 LOADED_DATA <= (others => '0');
27 else
28 if (LOAD = '1') then
29 LOADED_DATA <= DATA;
30 end if;
31 end if;
32 end process;
33
34 process
35 begin
36 wait until rising_edge(CLK);
37 if (RESET = '1') then
38 Q <= (others => '0');
39 TC <= '0';
40 else
41 if (Q < LOADED_DATA) then
42 Q <= Q + 1;
43 TC <= '0';
44 else
45 Q <= (others => '0');
46 TC <= '1';
47 end if;
48 end if;
49 end process;
50 end AAA;
```

הסימולציה



VIVADO DEVICE VIWE AND PIN LOCETION



Name	Direction	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type	Off-Chip Termination	IN_TERM
All ports (20)													
DATA[14]	IN		R2	☒	Multiple	LVC MOS33	3.300				NONE	NONE	
DATA[15]	IN		T1	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[16]	IN		U1	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[17]	IN		W2	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[18]	IN		R3	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[19]	IN		T2	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[20]	IN		T3	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[21]	IN		V2	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[22]	IN		W13	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[23]	IN		W14	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[24]	IN		V15	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[25]	IN		W15	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[26]	IN		W17	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[27]	IN		W16	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[28]	IN		V16	☒	34	LVC MOS33	3.300				NONE	NONE	
DATA[29]	IN		V17	☒	34	LVC MOS33	3.300				NONE	NONE	
Scalar ports (4)													
CLK	IN		W5	☒	34	LVC MOS33	3.300				NONE	NONE	
LOAD	IN		T17	☒	34	LVC MOS33	3.300				NONE	NONE	
RESET	IN		U18	☒	34	LVC MOS33	3.300				NONE	NONE	
TC	OUT		H1	☒	35	LVC MOS33	3.300		12	SLOW	NONE	FP_VTT_30	

XDC FILE

```
2
3 #####
4 # create clock #
5 #####
6
7 # Precision Generated
8 create_clock [get_ports {CLK}] -name {CLK} -period 10 -waveform {0 5}
9 create_clock -name {VirtualClock} -period 10 -waveform {0 5}
10
11 #####
12 # set input delay #
13 #####
14
15 # Precision Generated
16 set_input_delay 0 -clock [get_clocks {VirtualClock}] -add_delay [get_ports {DATA* LOAD RESET}]
17
18 #####
19 # set output delay #
20 #####
21
22 # Precision Generated
23 set_output_delay 0 -clock [get_clocks {VirtualClock}] -add_delay [get_ports {TC}]
24
25
26 #####
27 # Pin Locations and Voltages #
28 #####
29 set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCN0S33 } [get_ports { CLK }];
30 set_property -dict { PACKAGE_PIN U18 IOSTANDARD LVCN0S33 } [get_ports { RESET }];
31 set_property -dict { PACKAGE_PIN T17 IOSTANDARD LVCN0S33 } [get_ports { LOAD }];
32 set_property -dict { PACKAGE_PIN V17 IOSTANDARD LVCN0S33 } [get_ports { DATA[0] }];
33 set_property -dict { PACKAGE_PIN V16 IOSTANDARD LVCN0S33 } [get_ports { DATA[1] }];
34 set_property -dict { PACKAGE_PIN W16 IOSTANDARD LVCN0S33 } [get_ports { DATA[2] }];
35 set_property -dict { PACKAGE_PIN W17 IOSTANDARD LVCN0S33 } [get_ports { DATA[3] }];
36 set_property -dict { PACKAGE_PIN W15 IOSTANDARD LVCN0S33 } [get_ports { DATA[4] }];
37 set_property -dict { PACKAGE_PIN V15 IOSTANDARD LVCN0S33 } [get_ports { DATA[5] }];
38 set_property -dict { PACKAGE_PIN W14 IOSTANDARD LVCN0S33 } [get_ports { DATA[6] }];
39 set_property -dict { PACKAGE_PIN W13 IOSTANDARD LVCN0S33 } [get_ports { DATA[7] }];
40 set_property -dict { PACKAGE_PIN V2 IOSTANDARD LVCN0S33 } [get_ports { DATA[8] }];
41 set_property -dict { PACKAGE_PIN T3 IOSTANDARD LVCN0S33 } [get_ports { DATA[9] }];
42 set_property -dict { PACKAGE_PIN T2 IOSTANDARD LVCN0S33 } [get_ports { DATA[10] }];
43 set_property -dict { PACKAGE_PIN R3 IOSTANDARD LVCN0S33 } [get_ports { DATA[11] }];
44 set_property -dict { PACKAGE_PIN W2 IOSTANDARD LVCN0S33 } [get_ports { DATA[12] }];
45 set_property -dict { PACKAGE_PIN U1 IOSTANDARD LVCN0S33 } [get_ports { DATA[13] }];
46 set_property -dict { PACKAGE_PIN T1 IOSTANDARD LVCN0S33 } [get_ports { DATA[14] }];
47 set_property -dict { PACKAGE_PIN R2 IOSTANDARD LVCN0S33 } [get_ports { DATA[15] }];
48 set_property -dict { PACKAGE_PIN H1 IOSTANDARD LVCN0S33 } [get_ports { TC }];
49
50 #####
51 # Configuration Settings #
52 #####
53 set_property CFGBVS VCC0 [current_design]
54 set_property CONFIG_VOLTAGE 3.3 [current_design]
55
```

D6)

טווח התדר שנוכל לייצר על ידי הקלוק בעל תדירות ששווה ל

100MHz

הוא :

או אותו התדר (100 מגה הרץ)

או תדר קטן יותר שניתן על ידי ה

Data

הגדול ביותר

$$f = \frac{1}{2^{16} * 10 * 10^{-9}} = 1526Hz$$

לכן טווח התדרים של

100MHz > TC > 1526HZ